

Exam Guard



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1. Introduction

Examinations are critical assessments of a student's knowledge and skills. However, with the rise of Large Language Models (LLMs) such as ChatGPT and Gemini, students may attempt to bypass academic integrity by seeking instant answers online. **Exam Guard** is designed as a lightweight information security tool to help enforce exam discipline by restricting access to popular LLM websites through a user's hotspot.

2. Objectives

- Prevent connected devices from accessing LLM websites (e.g., ChatGPT, Gemini).
- Notify the hotspot owner immediately if a connected device attempts to bypass restrictions.
- Provide visibility into all devices connected to the hotspot, including their IP addresses.
- Enhance exam security in environments where internet access is allowed but misuse must be controlled.

3. System Features

- **Website Blocking:** Automatically blocks access to well-known LLM websites on all devices connected to the hotspot.
- **Security Breach Notification:** Displays a red blinking alert labeled "Security Breach Detected" whenever a blocked site is accessed.
- **Device Monitoring:** Shows the IP addresses of all connected devices.
- **Compatibility:** Works reliably with laptops, but has limitations with modern mobile phones (device names are not retrievable due to OS restrictions).

4. System Architecture

Exam Guard is built as a **multi-layered security system** that integrates network interception, device monitoring, and a real-time dashboard:

a. DNS Proxy (Port 53)

- Intercepts DNS queries from hotspot clients.
- Blocks AI domains by redirecting them to the instructor's IP.
- Non-blocked domains are forwarded to Google DNS (8.8.8.8).

b. HTTPS Interceptor (Port 443)

- Reads the TLS ClientHello SNI field to detect attempts to access blocked domains.

- Triggers alerts when violations occur.
- c. HTTP Interceptor (Port 80)**
 - Serves a custom “Access Blocked” page to students.
 - Logs the attempt for audit purposes.
- d. Flask Dashboard (Port 5000)**
 - Provides a SOC-style (Security Operations Center) interface.
 - Displays connected devices, their IP/MAC/hostname, and real-time alerts.
 - Uses SocketIO for instant push notifications.
- e. Background Scanner (ARP-based)**
 - Periodically scans the ARP table to detect connected devices.
 - Filters out virtual adapters and instructor’s own machine.
 - Resolves hostnames where possible (works reliably for laptops, limited for modern smartphones).
- f. Alert & Logging System**
 - Logs all violation attempts with timestamps.
 - Pushes critical alerts to the dashboard with device info.
 - Maintains a rolling log file (violations.txt) for later review.

5. Workflow

- **Setup Phase**
 - a) Instructor connects their laptop to the internet (Ethernet).
 - b) Windows Mobile Hotspot is enabled.
 - c) Exam Guard is launched with administrator privileges (python app.py).
 - d) The app auto-configures DNS routing.
- **Connection Phase**
 - a) Students connect their laptops to the hotspot.

- b) Exam Guard continuously monitors ARP to list connected devices (IP, MAC, hostname).
- **Monitoring & Blocking Phase**
 - a) When a student attempts to access a blocked AI domain:
 - 1) DNS proxy intercepts and redirects the query.
 - 2) HTTP interceptor serves the “Access Blocked” page.
 - 3) HTTPS interceptor detects SNI requests to blocked domains.
 - b) The system logs the attempt and triggers a **red blinking alert** on the dashboard.
- **Alerting Phase**
 - a) Dashboard displays:
 - 4) Device info (hostname/IP/MAC).
 - 5) Domain attempted.
 - 6) Timestamp of the violation.
 - b) Instructor sees a critical alert in real time.
- **Exit Phase**
 - a) On shutdown, Exam Guard restores original DNS settings.
 - b) Cleans up hosts file entries and firewall rules.
 - c) Logs remain available for post-exam review.

6. Wireframes

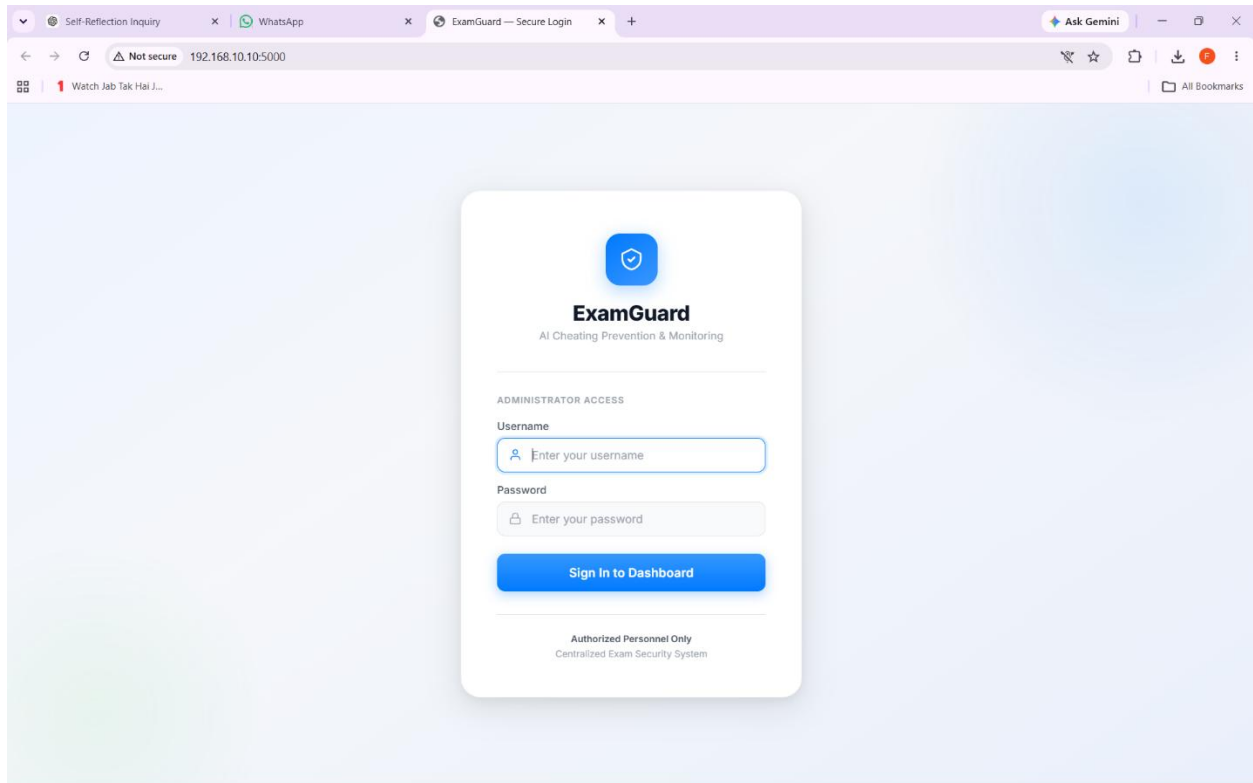


Figure 1: Login

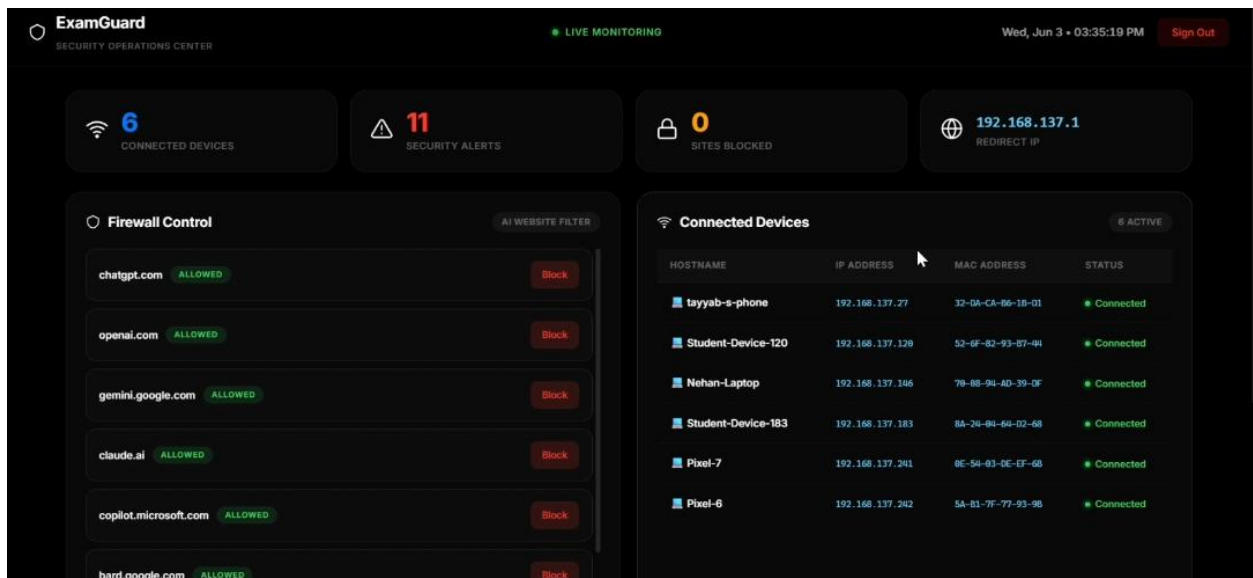


Figure 2: Dashboard

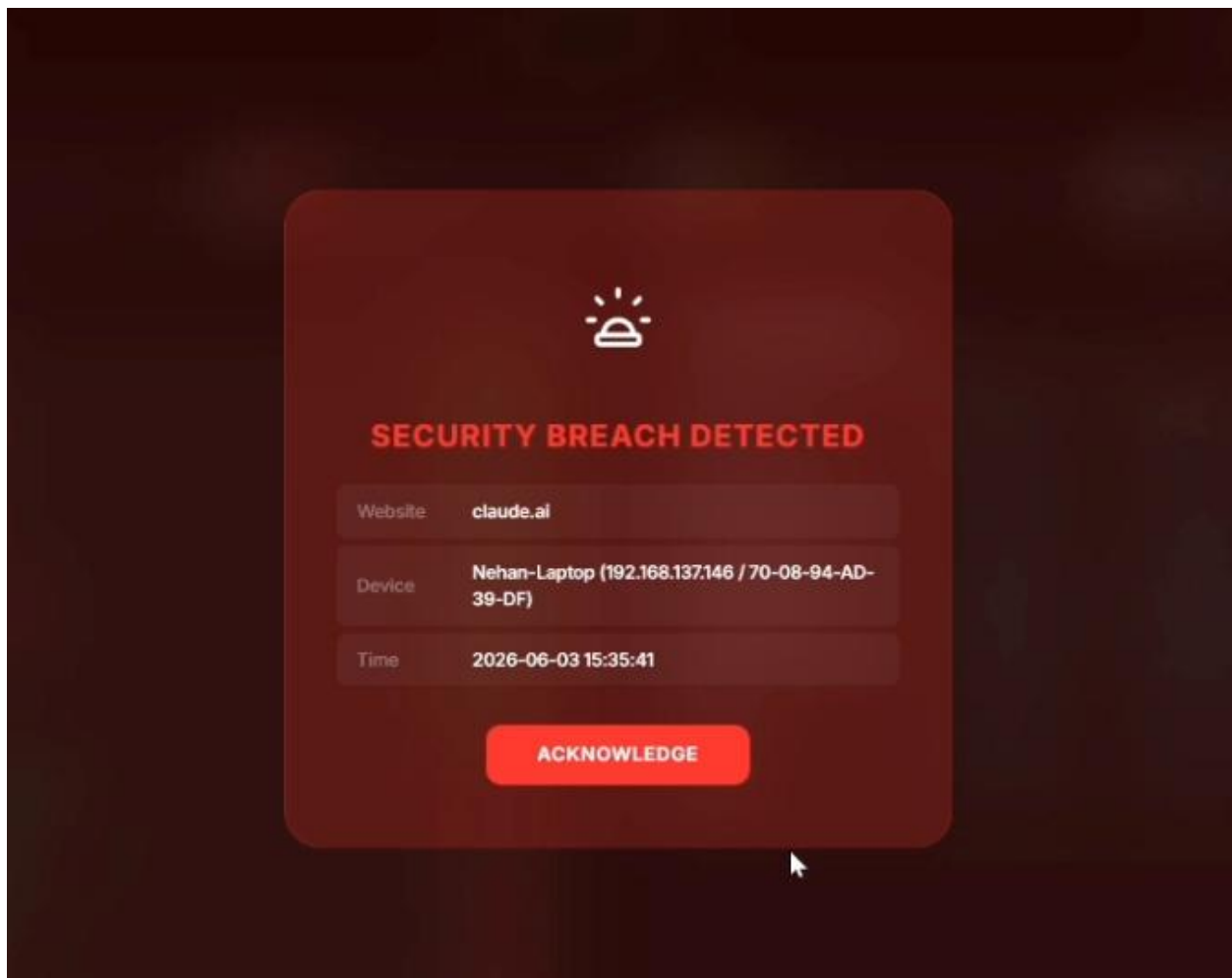


Figure 3: Security breach alert

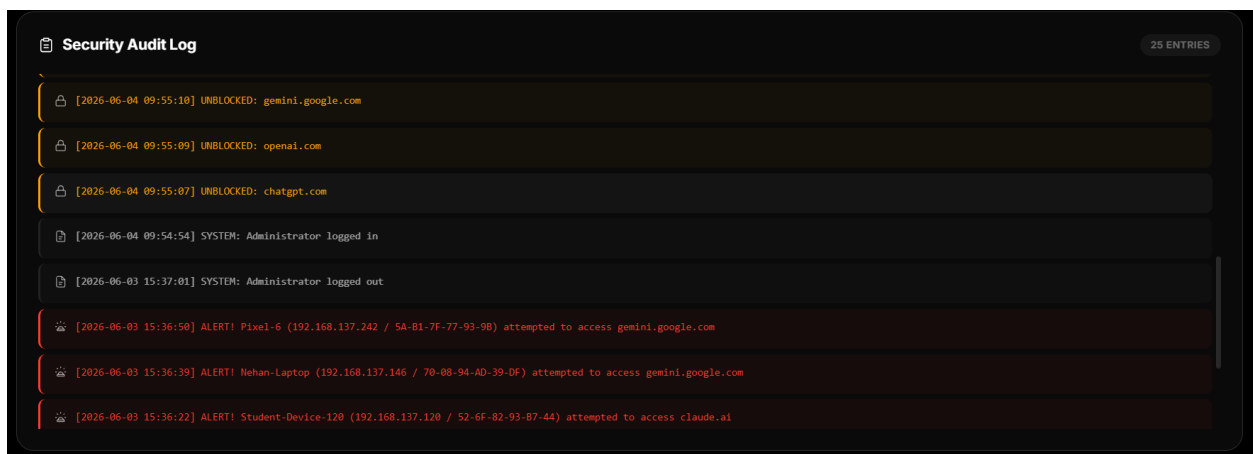


Figure 4: Security log

7. Use Cases

- **Classroom Exams:** Teachers can use Exam Guard to prevent students from accessing AI tools during online exams.
- **Group Study Sessions:** Ensures fair collaboration without reliance on external AI-generated answers.
- **Controlled Environments:** Useful in labs or workshops where internet access is necessary but restricted.

8. Limitations

- Cannot block mobile app versions of LLMs.
- Device names of modern smartphones are not visible due to OS-level privacy restrictions.
- Blocking is limited to hotspot traffic; devices using mobile data or VPNs can bypass restrictions.

9. Future Improvements

- Extend blocking to mobile applications.
- Improve device identification for modern smartphones.

10. Conclusion

Exam Guard is a practical solution for maintaining exam integrity in environments where internet access is required but misuse must be prevented. While it has limitations with mobile devices and apps, its ability to block LLM websites on laptops and provide real-time breach alerts makes it a valuable tool for educators and examiners.
